



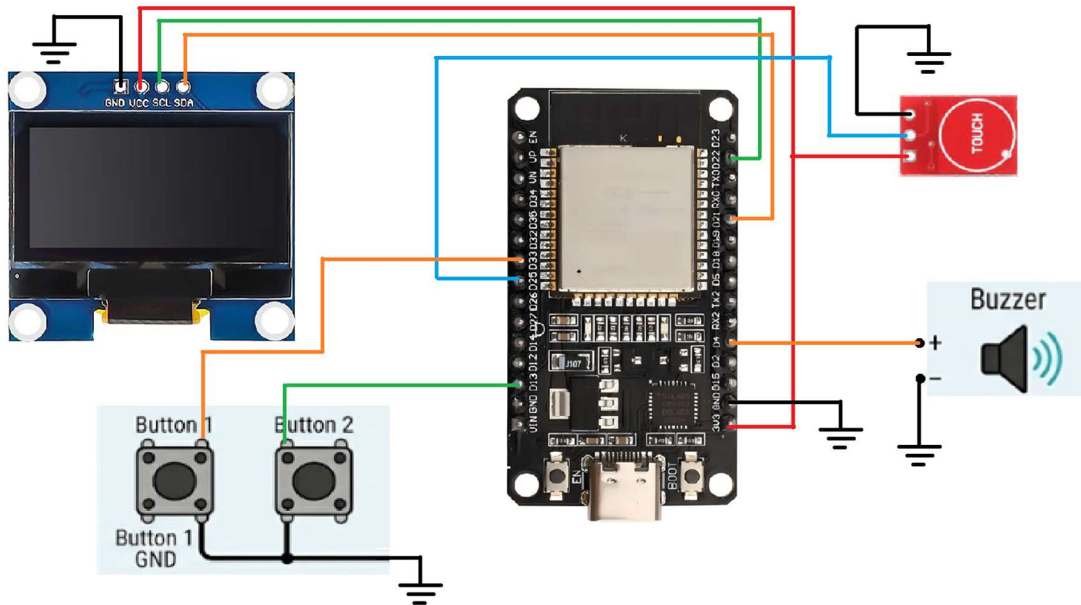
SYSTEM OVERVIEW

CYNO is designed as a compact and intelligent desk companion that integrates real-time notifications, visual output, and wireless communication through Wi-Fi. At its core, the system uses an ESP32 microcontroller, which manages key operations such as time synchronization, API communication, user input handling, and display control. The OLED display acts as the primary interface, presenting time, weather updates, notifications, and expressive robotic eye animations for interactive feedback. The system also includes a buzzer for alert functions and a button-based input mechanism for user interaction. CYNO combines embedded system design with IoT capabilities to deliver a responsive and user-friendly smart desk assistant.

SETUP

1. ESP32
2. OLED Display
3. Touch Sensor
4. Push Button
5. Buzzer Module
6. Wi-Fi Connectivity
7. Power Supply (USB)

CIRCUIT SETUP



CODE

CYNO CODE

HOW DOES THE SYSTEM WORK?

Once powered on, CYNO connects to Wi-Fi and synchronizes time using an NTP server. It continuously runs a web server to receive notification messages. When a message is received, the OLED display shows a blinking alert and the buzzer provides feedback.

The system operates in multiple modes:

- Clock mode displays time and date
- Uptime mode shows system runtime
- Weather mode fetches live data from an API

◆ User Interaction System

○ Button 1 (Mode)

- Single press → activates Uptime mode
- Double press → activates Weather mode
- During timer selection → used to start the timer

○ Button 2 (Timer Selection)

- Single press → selects 45-minute timer
- Double press → selects 10-minute timer

Timer ends → buzzer alert

Touch Sensor (Robotic Eye Control)

- Single touch → activates robotic eye (normal mode) 👁️
- Double touch → changes to angry mode 😡
- Hold for 3 seconds → switches to tired mode 😴

USEFUL TOOLS

1. Arduino IDE
2. ESP32 Board Package
3. Adafruit SSD1306 & GFX Libraries
4. ArduinoJson Library
5. HTTPClient Library
5. 3D Design Software (SolidWorks)

6. Slicing Software for 3D printing

SUMMARY

CYNO is a compact ESP32-based smart desk companion that integrates IoT features, real-time notifications, and interactive visual feedback. It demonstrates practical implementation of embedded systems, combining hardware, software, and design to create a functional and engaging desktop assistant.